

Script-Controlled ACL

Phase-Wise ServiceNow Lab Document

Purpose

This lab demonstrates how to implement a script-controlled Access Control List (ACL) to restrict users from viewing records unless they meet specific conditions. Only users belonging to the EEE branch are allowed to view records where the Branch field is set to EEE, while administrators retain full access.

Lab Duration	3 hrs
Custom Table	Institution Details (u_institution_details)
Security Model	Script-controlled READ + role-based CREATE/WRITE/DELETE

Phase Overview

Phase	Focus
1	Create test user and custom roles
2	Create Institution Details table and fields
3	Create records and configure READ ACL
4	Configure CREATE ACL
5	Configure WRITE ACL
6	Configure DELETE ACL
7	Final verification and outcome

Phase 1 – Creation of Users and Roles

Objective: Create the EEE test user, create the custom roles, and assign the roles to the test user.

1. Create the EEE User

1. Log in to the ServiceNow instance with admin access.
2. Navigate to User Administration → Users.
3. Click New.
4. User ID: EEE User.
5. First name: EEE.
6. Last name: User.
7. Email: eeeuser@gmail.com.
8. Save the record.

2. Create Custom Roles

1. Navigate to User Administration → Roles.
2. Click New.
3. Name: bb1.
4. Description: Brief role purpose.
5. Click Submit.
6. Create the other roles bb2, bb3, and bb4 using the same procedure.
7. Assign all four roles to EEE User.

Phase 2 – Create the Institution Details Table

Objective: Create the custom table that will be protected by the ACLs.

1. Create the Table

1. In the Application Navigator, search for System Definition → Tables.
2. Click New.
3. Label: Institution Details.
4. Name: u_institution_details.
5. Extends for: false.
6. Click Submit.

2. Add Table Fields

Field	Type
Student Roll Number	Auto Number
Student Name	Reference – User
Faculty Name	Reference – User
Branch	Choice (ECE, EEE, CSE)
Email	String
Phone Number	String
Description	Multi String

Save the table after adding the fields.

Phase 3 – Create Records and Configure READ ACL

Objective: Create sample records with different branch values and restrict READ access so that EEE users can access EEE records.

1. Create Sample Records

1. Create multiple records in the Student Records table with different branch values: ECE, EEE, and CSE.

2. Create the READ ACL

1. In the Application Navigator, search for System Security → Access Control (ACL).
2. Elevate the Security_admin role.
3. Click New to create a new ACL.
4. Type: record.
5. Operation: read.
6. Name: u_institution_details.
7. Active: true.
8. Advanced: true.
9. In the Requires role related list, add bb1.
10. Add Data condition: Branch is EEE.
11. Add the script below.
12. Click Submit to save the ACL.

READ ACL Script

```
(function () { // Allow admin users full access if (gs.hasRole('admin')) { return true; } //  
Allow only EEE branch users to see EEE records if (gs.hasRole('bb1')){ return true; } // Deny  
access for all others return false; })();
```

3. Verify READ Access

- Impersonate EEE users and open the Student Records list.
- Copy the table name u_institution_details.list/LIST and paste it in the All menu.
- Impersonate Admin and open the Student Records list.
- User with bb1 role can view only EEE branch records.
- User without the role cannot view any records.
- Admin user can view all records regardless of branch.

Phase 4 – Configure CREATE ACL

Objective: Control creation of records using the bb2 role.

1. Create the CREATE ACL

1. Click New to create a new ACL (Create).
2. Type: record.
3. Operation: create.
4. Name: u_institution_details.
5. Active: true.
6. In the Requires role related list, add bb2.
7. Do not add a Data condition.
8. Save the ACL record.

2. Verify CREATE Access

A user with bb1 and bb2 roles can view only EEE branch records and has the New button available.

Phase 5 – Configure WRITE ACL

Objective: Allow editing through the bb3 role in addition to the existing access requirements.

1. Create the WRITE ACL

1. Click New to create a new ACL (Write).
2. Type: record.
3. Operation: Write.
4. Name: u_institution_details.
5. Active: true.
6. In the Requires role related list, add bb3.
7. Do not add a Data condition.
8. Save the ACL record.

2. Verify WRITE Access

A user with bb1, bb2, and bb3 roles can view and edit the EEE branch records along with the New button.

Phase 6 – Configure DELETE ACL

Objective: Allow deletion through the bb4 role after the required role chain is satisfied.

1. Create the DELETE ACL

1. Click New to create a new ACL (Delete).
2. Type: record.
3. Operation: Delete.
4. Name: u_institution_details.
5. Active: true.
6. In the Requires role related list, add bb4.
7. Do not add a Data condition.
8. Save the ACL record.

2. Verify DELETE Access

Users with bb1, bb2, bb3, and bb4 roles can see, create, edit, and delete the EEE branch records.

Phase 7 – Final Verification and Outcome

Final Access Model

Operation	Required role	Verification
READ	bb1	EEE branch records only; admin can access all
CREATE	bb2	New button available with required role chain
WRITE	bb3	EEE records can be edited with required role chain
DELETE	bb4	EEE records can be deleted with required role chain

Outcome

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.

Completion Checklist

- EEE User created and saved.
- Roles bb1, bb2, bb3, and bb4 created.
- All roles assigned to EEE User.
- u_institution_details table created.
- Required fields added.
- Sample ECE, EEE, and CSE records created.
- READ ACL configured with bb1, Branch is EEE, and script.
- CREATE ACL configured with bb2.
- WRITE ACL configured with bb3.
- DELETE ACL configured with bb4.
- Access verified for the required role combinations and Admin.